L06: Estimation with Structural Models

S&DS 6165: Topics in Causal Inference

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2026-10-28

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Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Use a restriction on how treatments interact to learn about a full rollout from a partially treated network.
- Derive an estimator first for additive spillovers, then for low-order interactions, using the experiment's known assignment probabilities.
- Connect this construction to least squares and explain which treatment contrasts the design can recover.
- Examine precision, model error, clustered designs, and covariate adjustment.
- Contrast low-order assumptions with monotonicity, which restricts the direction of spillovers rather than their interaction order.

1 Learning about a rollout before everyone is treated

Suppose a platform is testing a feature that makes it easier for users to send messages. A user's activity may increase because they receive the feature, because their friends receive it, or because a particular combination of friends receives it. The policy question concerns a full rollout: how would average activity change if everyone received the feature? The experiment, however, treats only a random subset of users.

L05 approached this problem by describing which assignments give a unit the same outcome as the all-treated or all-control assignment. Under neighborhood interference, observing every member of a unit's neighborhood in treatment provides its all-treated outcome. With independent treatment probability p, this happens with probability p^{d_i}, where d_i is the number of units in that neighborhood. These observations can be scarce even when each unit has a reasonable chance of treatment.

There is another source of information. If each friend's contribution is additive, a neighborhood containing a mixture of treated and untreated users still reveals something about the full rollout. We need not wait until all friends receive treatment simultaneously to learn their combined contribution. If interactions occur only among pairs of friends, we need information about pairs, but not an unrestricted description of every possible neighborhood assignment.

This lecture develops that idea through the low-order outcome framework of Cortez-Rodriguez, Eichhorn, and Yu (2023). Its central move is to restrict how many treatment indicators can interact in an outcome. That restriction makes a global treatment effect a linear function of a manageable collection of coefficients. The assignment design then supplies the equations needed to estimate that linear function.

The reasoning remains design-based. The network, potential outcomes, and coefficients are fixed. Treatment assignment is random. A structural model is used to restrict the fixed potential-outcome schedule; it does not introduce random regression errors or a population of interchangeable units.

2 A model of interactions, rather than a model of identical units

2.1 Setup and target

There are N experimental units. Write $\mathbf{z} = (z_1, \dots, z_N) \in \{0,1\}^N$ for a possible assignment and $\mathbf{Z}$ for the random assignment. Unit i has potential outcome $Y_i(\mathbf{z})$ and observed outcome $Y_i^{\text{obs}} = Y_i(\mathbf{Z})$.

Let $\mathcal{N}_i$ be the known set of units whose assignments can affect unit i's outcome, including i itself. We call it a closed interference neighborhood and set $d_i = |\mathcal{N}_i|$. For a directed network, membership $j \in \mathcal{N}_i$ means that j can influence i; the reverse need not hold. Neighborhood interference means that assignments outside $\mathcal{N}_i$ do not affect Y_i. A neighborhood may be a conservative superset of the actual influencing units. Missing an influencing unit is a different matter: that can invalidate the model.

The target is the **global average treatment effect** (GATE), also called the total treatment effect (TTE):

$$\tau_{\text{G}} = \frac{1}{N}\sum_{i=1}^{N}\{Y_i(\mathbf{1}_N) - Y_i(\mathbf{0}_N)\}. \tag{1}$$

Here $\mathbf{1}_N$ and $\mathbf{0}_N$ assign everyone to treatment and control, respectively. This effect combines direct and spillover effects. It does not hold neighbors' treatments fixed while changing only a unit's own treatment.

Initially, assignments are independent, with $Z_j \sim \text{Bernoulli}(\pi_j)$ and known $0 < \pi_j < 1$. We use $\mathbb{E}_Z$ and Var_Z for expectation and variance under this design, keeping the entire potential-outcome schedule fixed. Common-probability assignment is the special case $\pi_j = p$.

2.2 Every binary response surface has a polynomial representation

For a two-person neighborhood $\mathcal{N}_i = \{j, k\}$, where one of j, k is i itself, the outcome has four possible values. They can always be written as

$$Y_i(z_j, z_k) = c_{i,\emptyset} + c_{i,\{j\}} z_j + c_{i,\{k\}} z_k + c_{i,\{j,k\}} z_j z_k. \tag{2}$$

The intercept is the outcome when neither person is treated. The first-order coefficient $c_{i,\{j\}}$ is the effect of treating j when k is untreated. The pair coefficient is the additional effect of treating both people beyond the sum of their effects when treated separately. Explicitly,

$$\begin{aligned}
c_{i,\emptyset} &= Y_i(0,0),\\
c_{i,\{j\}} &= Y_i(1,0) - Y_i(0,0),\\
c_{i,\{k\}} &= Y_i(0,1) - Y_i(0,0),\\
c_{i,\{j,k\}} &= Y_i(1,1) - Y_i(1,0) - Y_i(0,1) + Y_i(0,0).
\end{aligned}$$

The interaction coefficient is a difference in the effect of j between the two treatment states of k. It can be positive, negative, or zero. Thus a polynomial here is not an approximation to a smooth

curve on a continuous domain. With binary inputs, it is an exact way to organize the entries of the science table.

More generally, every function on a binary neighborhood has a unique multilinear representation

$$Y_i(\mathbf{z}) = \sum_{S \subseteq \mathcal{N}_i} c_{i,S} \prod_{j \in S} z_j. \tag{3}$$

The product over the empty set is defined to equal one. Multilinear means that no indicator appears with a power greater than one: for binary treatment, $z_j^2 = z_j$, so higher powers add no information. There are 2^{d_i} coefficients, matching the number of possible local assignments. The representation itself therefore imposes no restriction beyond neighborhood interference.

To see uniqueness, evaluate the function at the assignment treating exactly the members of some subset $T \subseteq \mathcal{N}_i$. Its value is $\sum_{S \subseteq T} c_{i,S}$. Starting with $T = \emptyset$, then the singletons, pairs, and successively larger sets, each new equation identifies one new coefficient after subtracting the coefficients already determined.

2.3 Low order is the substantive restriction

The **order-β model** imposes

$$c_{i,S} = 0 \quad \text{whenever } |S| > \beta. \tag{4}$$

Define $\mathcal{S}_i^\beta = \{S \subseteq \mathcal{N}_i : |S| \leq \beta\}$, including the empty set. The model becomes

$$Y_i(\mathbf{z}) = \sum_{S \in \mathcal{S}_i^\beta} c_{i,S} \prod_{j \in S} z_j. \tag{5}$$

For $\beta = 1$, all influences are additive. A unit can have many influencing neighbors, and their effects can all differ. For $\beta = 2$, any pair of assignments can interact, but the contribution of that pair cannot depend on a third assignment. Neither specification requires common coefficients across units. The effect of j on i may differ from its effect on k, and direct effects may differ throughout the network.

Low order consequently differs from low degree in the network. A user with hundreds of friends can satisfy an additive model. Conversely, an outcome that changes only when all three members of a small neighborhood receive treatment requires a third-order term. It also differs from assuming that only the fraction of treated neighbors matters: the polynomial allows different friends to matter differently.

The number of retained coefficients for unit i is $K_i = \sum_{r=0}^{\min(\beta, d_i)} \binom{d_i}{r}$. For fixed small β, this grows polynomially with neighborhood size, rather than as 2^{d_i}. But choosing a small order is a scientific assumption, not merely a way to reduce computation. A threshold or saturation response need not have small degree just because its verbal description is simple.

2.4 The global effect is a sum of coefficients

At the all-treated assignment, every product in Equation 5 equals one. At the all-control assignment, only the empty-set term remains. Therefore,

$$Y_i(\mathbf{1}_N) - Y_i(\mathbf{0}_N) = \sum_{\substack{S \in \mathcal{S}_i^\beta \\ S \neq \emptyset}} c_{i,S}, \qquad \tau_{\mathrm{G}} = \frac{1}{N} \sum_i \sum_{\substack{S \in \mathcal{S}_i^\beta \\ S \neq \emptyset}} c_{i,S}. \tag{6}$$

The intercept cancels, but every nonempty interaction enters the global contrast. We have converted the causal target into a linear function of the coefficients. We next ask whether the experiment can estimate that function without learning each response surface accurately.

3 Start with additive spillovers

3.1 Randomization isolates a neighbor's contribution

Under $\beta = 1$, write the response as

$$Y_i(\mathbf{Z}) = c_{i,\emptyset} + \sum_{k \in \mathcal{N}_i} c_{i,\{k\}} Z_k.$$

To learn the contribution of a particular neighbor j, multiply the observed outcome by the centered treatment indicator $Z_j - \pi_j$. Independence makes this indicator uncorrelated with the intercept and with every other neighbor's assignment. The one surviving term is

$$\begin{aligned} \mathbb{E}_Z[Y_i^{\mathrm{obs}}(Z_j - \pi_j)] &= c_{i,\emptyset} \mathbb{E}_Z[Z_j - \pi_j] + \sum_{k \in \mathcal{N}_i} c_{i,\{k\}} \mathbb{E}_Z[Z_k(Z_j - \pi_j)] \\ &= c_{i,\{j\}} \pi_j (1 - \pi_j). \end{aligned} \tag{7}$$

Dividing by the treatment variance gives an unbiased estimator of $c_{i,\{j\}}$. Summing these estimates over j and then averaging over outcome units yields

$$\hat{\tau}_{\mathrm{SNIPE}(1)} = \frac{1}{N} \sum_i Y_i^{\mathrm{obs}} \sum_{j \in \mathcal{N}_i} \frac{Z_j - \pi_j}{\pi_j (1 - \pi_j)}. \tag{8}$$

SNIPE stands for Structured Neighborhood Interference Polynomial Estimator (Cortez-Rodriguez, Eichhorn, and Yu 2023). Each outcome receives a weight determined by the assignments of all units that can influence it. For additive interference, the weight accumulates the evidence supplied by each neighbor's randomized assignment.

This expression has the familiar Horvitz–Thompson estimator as a special case. If $\mathcal{N}_i = \{i\}$, then

$$\frac{Z_i - \pi_i}{\pi_i (1 - \pi_i)} = \frac{Z_i}{\pi_i} - \frac{1 - Z_i}{1 - \pi_i}.$$

Under interference, observing Y_i gives information about effects associated with several assignments, not just Z_i. The extra terms use that information under the additive restriction.

3.2 An explicit four-assignment calculation

Consider one outcome with neighborhood $\{1, 2\}$ and response $Y_i(z_1, z_2) = 10 + 2z_1 + 3z_2$. The global contrast for this unit is 5. Under independent fair coins, each of the four assignments occurs with probability 1/4. Let ℓ_i denote the inner sum of weights in Equation 8. The contribution from this outcome is $Y_i^{\text{obs}}\ell_i$.

(Z_1, Z_2)	Y_i^{obs}	$\ell_i = 4(Z_1 - 1/2) + 4(Z_2 - 1/2)$	$Y_i^{\text{obs}}\ell_i$
$(0, 0)$	10	-4	-40
$(1, 0)$	12	0	0
$(0, 1)$	13	0	0
$(1, 1)$	15	4	60

The average of the last column is 5. A single contribution can nevertheless be far from 5. Unbiasedness does not mean that one observed outcome estimates all of its coefficients precisely. Precision, if available, comes from averaging contributions across many units while controlling their dependence.

The intercept of 10 contributes random noise although it cancels from the target. Subtracting a known baseline would leave the expectation unchanged and could improve precision. We return to this observation when introducing covariate adjustment.

With two neighbors and fair coins, the mixed assignments happen to receive zero weight. This cancellation is special. With three neighbors and fair coins, the weight is $4\sum_{j\in\mathcal{N}_i} Z_j - 6$, which equals -2 or 2 for mixed assignments containing one or two treated units. Those assignments directly contribute to the estimator.

4 From coefficient estimation to the pseudoinverse

4.1 The known design replaces a missing regression sample

The additive calculation used independence to isolate one coefficient. A matrix formulation will handle interactions and more general designs with the same underlying logic.

Choose an ordering of $\mathcal{S}_i^\beta$, putting the empty set first. Let $\mathbf{v}_i(\mathbf{z})$ be the K_i-dimensional column vector whose entry indexed by S is $\prod_{j\in S} z_j$. Let $\mathbf{c}_i$ collect the coefficients in the same order. For example, with two neighbors and $\beta = 2$,

$$\mathbf{v}_i(\mathbf{z}) = (1, z_1, z_2, z_1 z_2)^\top, \qquad Y_i(\mathbf{z}) = \mathbf{v}_i(\mathbf{z})^\top \mathbf{c}_i.$$

The vector $\mathbf{a}_i = \mathbf{v}_i(\mathbf{1}_N) - \mathbf{v}_i(\mathbf{0}_N)$ has first entry zero and all remaining entries one. It specifies the desired coefficient sum: the unit's global contrast is $\mathbf{a}_i^\top \mathbf{c}_i$.

Now imagine, only as a thought experiment, repeating the assignment many times for the same unit with its response function unchanged. With enough repeated assignments, ordinary least squares would regress the repeated outcomes on the corresponding feature vectors. The population normal equations would be

$$\underbrace{\mathbb{E}_Z[\mathbf{v}_i(\mathbf{Z})\mathbf{v}_i(\mathbf{Z})^\top]}_{\mathbf{M}_i}\,\mathbf{c}_i = \mathbb{E}_Z[\mathbf{v}_i(\mathbf{Z})Y_i^{\text{obs}}]. \tag{9}$$

The $K_i \times K_i$ matrix $\mathbf{M}_i$ is the **design moment matrix**. Its entries depend on assignment probabilities, not unknown outcome coefficients. For Bernoulli assignment,

$$(\mathbf{M}_i)_{S,T} = \mathbb{E}_Z\left[\prod_{j\in S\cup T} Z_j\right] = \prod_{j\in S\cup T}\pi_j. \tag{10}$$

The union appears because an indicator present in both products still equals itself when squared. The first entry, indexed by two empty sets, equals one.

In the actual experiment we have one outcome per unit, not a sample of repeated outcomes for each unit. We therefore cannot run a separate ordinary regression and identify all of its coefficients from that one row. Nor can we silently pool units: their coefficient vectors are allowed to differ.

What we can do is use the known left-hand matrix in Equation 9 and an unbiased, one-observation estimate of its right-hand side. When $\mathbf{M}_i$ is invertible,

$$\hat{\mathbf{c}}_i = \mathbf{M}_i^{-1}\mathbf{v}_i(\mathbf{Z})Y_i^{\text{obs}}, \qquad \mathbb{E}_Z[\hat{\mathbf{c}}_i] = \mathbf{c}_i. \tag{11}$$

The apparent abundance of unbiased coefficient estimates is not an abundance of information in one realized assignment. All entries of $\hat{\mathbf{c}}_i$ use the same outcome and can be very noisy and strongly dependent. Their expectations recover the coefficients across assignments.

4.2 What singularity means

Independent Bernoulli assignment with every probability strictly between zero and one gives positive probability to every local assignment. The distinct multilinear features are linearly independent on that support, so $\mathbf{M}_i$ is positive definite. Other designs can make it singular. For example, always assigning two neighbors together makes their two treatment indicators identical in every experiment.

The Moore–Penrose pseudoinverse, written $\mathbf{M}_i^\dagger$, inverts the nonzero eigenvalues of this symmetric matrix and leaves its zero eigenvalues at zero. The associated estimator is

$$\hat{\tau}_{\text{PI}} = \frac{1}{N}\sum_i Y_i^{\text{obs}}\ell_i(\mathbf{Z}), \qquad \ell_i(\mathbf{Z}) = \mathbf{a}_i^\top\mathbf{M}_i^\dagger\mathbf{v}_i(\mathbf{Z}). \tag{12}$$

PI denotes pseudoinverse. This is the pseudoinverse of an expectation under the design, not the pseudoinverse of the single realized outer product $\mathbf{v}_i(\mathbf{Z})\mathbf{v}_i(\mathbf{Z})^\top$.

Proposition 4.1 (When the pseudoinverse estimates the desired contrast)**.** *Suppose the order-β response model holds. If $\mathbf{a}_i$ lies in the column space of $\mathbf{M}_i$ for every i, then $\mathbb{E}_Z[\hat{\tau}_{\mathrm{PI}}] = \tau_{\mathrm{G}}$. Because $\mathbf{M}_i$ is symmetric, its column and row spaces coincide.*

Proof. For each unit, substitute $Y_i^{\mathrm{obs}} = \mathbf{v}_i(\mathbf{Z})^\top \mathbf{c}_i$ to obtain

$$\begin{aligned}\mathbb{E}_Z[Y_i^{\mathrm{obs}} \ell_i(\mathbf{Z})] &= \mathbf{a}_i^\top \mathbf{M}_i^\dagger \mathbb{E}_Z[\mathbf{v}_i(\mathbf{Z})\mathbf{v}_i(\mathbf{Z})^\top]\mathbf{c}_i \\ &= \mathbf{a}_i^\top \mathbf{M}_i^\dagger \mathbf{M}_i \mathbf{c}_i.\end{aligned}$$

The matrix $\mathbf{P}_i = \mathbf{M}_i^\dagger \mathbf{M}_i$ is the orthogonal projection onto the column space of $\mathbf{M}_i$. The stated condition gives $\mathbf{a}_i^\top \mathbf{P}_i = \mathbf{a}_i^\top$, so the expectation is $\mathbf{a}_i^\top \mathbf{c}_i$. Average these equalities over units. □

Singularity can thus prevent estimation of individual coefficients without preventing estimation of their sum. If two neighbors always move together, their separate additive effects cannot be distinguished, but their combined effect may remain estimable. The target contrast, not the full coefficient vector, determines the required support condition.

The condition is also necessary for unbiased estimation of every possible coefficient vector in this unrestricted unit-specific model. If it fails, there is a vector $\mathbf{b}_i$ in the null space of $\mathbf{M}_i$ for which $\mathbf{a}_i^\top \mathbf{b}_i \neq 0$. Since

$$0 = \mathbf{b}_i^\top \mathbf{M}_i \mathbf{b}_i = \mathbb{E}_Z[(\mathbf{v}_i(\mathbf{Z})^\top \mathbf{b}_i)^2],$$

adding $\mathbf{b}_i$ to the coefficients changes no outcome on the design's support. It does change the desired global contrast. The experiment cannot distinguish these two response schedules. Applying a pseudoinverse does not resolve this identification failure; it selects the projection that is visible under the design.

4.3 The geometry behind the weights

The estimation problem can also be expressed without estimating coefficients. We want a random weight $w_i(\mathbf{Z})$ that reproduces the global contrast for every allowed polynomial response. That requires the coordinatewise moment equations

$$\mathbb{E}_Z[w_i(\mathbf{Z})\mathbf{v}_i(\mathbf{Z})] = \mathbf{a}_i. \tag{13}$$

The first equation says the weight has mean zero, so a constant outcome makes no contribution in expectation. Every other equation says that a permitted treatment product receives the correct coefficient of one in the target.

When the estimability condition holds, these equations are feasible. Among weights satisfying them, ℓ_i in Equation 12 has minimum $\mathbb{E}_Z[w_i^2]$. Indeed, if $w_i = \ell_i + r_i$, the moment equations give $\mathbb{E}_Z[r_i \mathbf{v}_i] = 0$. Since ℓ_i is a linear combination of these features, $\mathbb{E}_Z[r_i \ell_i] = 0$, and

$$\mathbb{E}_Z[w_i^2] = \mathbb{E}_Z[\ell_i^2] + \mathbb{E}_Z[r_i^2].$$

This is a least-squares projection argument: the additional component of a weight orthogonal to the allowed response features cannot help reproduce their contrast. Minimum squared weight is useful for controlling noise, but it is not a universal minimum-variance claim for the whole estimator. Outcomes differ across assignments, and contributions from different units can covary.

5 Higher-order interactions under independent assignment

5.1 Centering provides an orthogonal basis

The raw monomial matrix can be large. Independent assignment gives a more transparent route to the same estimator. For a subset S of the neighborhood, define the centered product

$$u_S(\mathbf{Z}) = \prod_{j \in S} (Z_j - \pi_j), \qquad u_\emptyset(\mathbf{Z}) = 1.$$

If $S \neq T$, some indicator occurs in exactly one of the products. Its mean-zero factor and independence imply $\mathbb{E}_Z[u_S u_T] = 0$. Moreover, $\mathbb{E}_Z[u_S^2] = \prod_{j \in S} \pi_j(1 - \pi_j)$. These centered products span the same order-β polynomial space as the raw products. Subtracting constants from the inputs changes coefficients, but not the maximum degree.

We can therefore write $Y_i(\mathbf{z}) = \sum_{S \in \mathcal{S}_i^\beta} b_{i,S} u_S(\mathbf{z})$, where the $b_{i,S}$ are coefficients in this centered basis. Orthogonality gives

$$\mathbb{E}_Z \left[Y_i^{\text{obs}} \frac{u_S(\mathbf{Z})}{\prod_{j \in S} \pi_j(1 - \pi_j)} \right] = b_{i,S}.$$

The global contrast of the basis function u_S is

$$\Delta_S = u_S(\mathbf{1}_N) - u_S(\mathbf{0}_N) = \prod_{j \in S} (1 - \pi_j) - \prod_{j \in S} (-\pi_j).$$

For the empty set this difference is zero. Combining the unbiased estimates of the centered coefficients with their global contrasts gives

$$\hat{\tau}_{\text{SNIPE}(\beta)} = \frac{1}{N} \sum_i Y_i^{\text{obs}} \sum_{\substack{S \in \mathcal{S}_i^\beta \\ S \neq \emptyset}} \Delta_S \prod_{j \in S} \frac{Z_j - \pi_j}{\pi_j(1 - \pi_j)}. \tag{14}$$

This explicit formula equals Equation 12 under independent Bernoulli assignment. The centered basis diagonalizes the design inner product, doing the work that matrix inversion performs in the raw basis.

5.2 A pair interaction and a revealing cancellation

Consider again a two-person neighborhood, now with $Y_i(z_1, z_2) = 10 + 2z_1 + 3z_2 + 4z_1z_2$. Its global contrast is 9. Take common treatment probability $p = 1/4$ and define $q_j = (Z_j - p)/\{p(1-p)\}$. Each singleton has $\Delta_{\{j\}} = 1$, while $\Delta_{\{1,2\}} = (1-p)^2 - p^2 = 1 - 2p = 1/2$. The order-two weight is

$$\ell_i = q_1 + q_2 + \tfrac{1}{2}q_1q_2.$$

The first two terms alone recover the additive contribution 5 plus $2p \cdot 4 = 2$ from the pair interaction. To check the latter statement, $\mathbb{E}_Z[Z_1Z_2q_1] = p$ and $\mathbb{E}_Z[Z_1Z_2q_2] = p$. The pair term supplies the missing amount: $\mathbb{E}_Z[Z_1Z_2q_1q_2] = 1$, so its contribution is $(1/2) \cdot 4 = 2$. The expectation of the full contribution is therefore $5 + 2 + 2 = 9$.

This calculation also explains why the additive estimator is generally biased when pair interactions are present. It estimates only the part of the pair effect that appears when the other member is treated at the experiment's assignment probability. The full rollout requires that other member to be treated with certainty.

There is an instructive exception. At $p = 1/2$, the pair contrast $\Delta_{\{1,2\}}$ is zero, so the order-two and order-one SNIPE weights coincide. Pair effects still enter the GATE. They are already captured through the first-order centered coefficients: changing from raw to centered products moves part of a raw pair term into the linear terms. For the example above, the additive weight at $p = 1/2$ recovers $5 + 2p \cdot 4 = 9$.

This cancellation does not make arbitrary high-order interactions harmless. For a raw triple term $\gamma z_1 z_2 z_3$, the order-one weight at common probability p has expectation $3p^2\gamma$, rather than γ. At $p = 1/2$ it recovers only $3\gamma/4$. Correctness depends jointly on the target, the response model, and the design.

6 Precision requires information about shared assignments

6.1 Interaction order controls a local difficulty

The centered representation makes the squared size of a unit's weight explicit:

$$\mathbb{E}_Z[\ell_i^2] = \sum_{\substack{S \in \mathcal{S}_i^\beta \\ S \neq \emptyset}} \frac{\Delta_S^2}{\prod_{j \in S} \pi_j(1-\pi_j)}. \tag{15}$$

Products of small treatment probabilities still matter. However, the largest product now involves at most β units, rather than the entire neighborhood. For fixed β and probabilities bounded away from zero and one, the number of terms grows polynomially with d_i. This is the mechanism by which low order can improve precision relative to requiring full neighborhood exposure. It does not guarantee small variance for every network or every outcome schedule.

For additive interference with common p, the formula simplifies to $\mathbb{E}_Z[\ell_i^2] = d_i/\{p(1-p)\}$. Each extra influencing assignment adds to the random fluctuation of the weight. A high-degree outcome can be informative about many treatment contributions, but that information is not costless.

6.2 Shared neighbors generate covariance

Let $T_i = Y_i^{\text{obs}}\ell_i(\mathbf{Z})$ be unit i's contribution. The exact variance is

$$\text{Var}_Z(\hat{\tau}_{\text{PI}}) = \frac{1}{N^2}\sum_{i,k}\text{Cov}_Z(T_i, T_k). \tag{16}$$

Under independent unit assignment and neighborhood interference, T_i and T_k are independent when $\mathcal{N}_i \cap \mathcal{N}_k = \emptyset$. When their neighborhoods overlap, one treatment coin can affect both weighted outcomes. The number of units is therefore not the number of independent pieces of information.

A simple bound exposes this distinction. Suppose $|Y_i(\mathbf{z})| \leq B$ for all units and assignments, let $\mathbb{E}_Z[\ell_i^2] \leq A$, and suppose each neighborhood overlaps with at most D neighborhoods, counting itself. By Cauchy–Schwarz, $|\text{Cov}_Z(T_i, T_k)| \leq B^2A$. There are at most ND potentially nonzero covariance terms, so

$$\text{Var}_Z(\hat{\tau}_{\text{PI}}) \leq \frac{B^2AD}{N}. \tag{17}$$

This deliberately elementary bound separates three difficulties: outcome magnitude, the size of the weights, and dependence across contributions. It also shows why bounding neighborhood size alone is insufficient in a directed network. A single influential unit may appear in many small neighborhoods, creating widespread dependence.

Cortez-Rodriguez, Eichhorn, and Yu (2023) give sharper variance bounds and asymptotic normality results under their network and outcome regularity conditions. Such results require a sequence of experiments in which dependence and influential contributions remain controlled. Exact unbiasedness by itself supplies neither a central limit theorem nor a reliable standard error.

6.3 Inference is a further problem

An empirical variance treating the T_i as independent drops the covariance terms in Equation 16. A usual regression standard error does not recover those terms either. The relevant dependence comes from shared assignments, and uncertainty calculations must reflect it.

Even with a known design, exact variance estimation is difficult. The variance contains products of unknown potential-outcome coefficients, while each unit reveals only one outcome. Unbiased estimation of each coefficient does not imply that multiplying two estimated coefficients gives an unbiased estimate of their product. Conservative bounds and dependence-aware asymptotic variance estimators address different parts of this difficulty. A normal approximation should be paired with a justified variance procedure and the assumptions under which the approximation is valid.

For a small network, exact enumeration is useful for checking an implementation under a specified response schedule. It is a simulation exercise: the missing potential outcomes are supplied by the analyst. It does not turn the variance under the real, unknown schedule into an observed quantity.

7 Changing the design while retaining the model

The moment equations did not require independent treatment assignment. What changes under a different design is $\mathbf{M}_i$, its column space, and the covariance between contributions. Eichhorn, Khan, Ugander, and Yu (2026) develop this general-design perspective, with particular attention to clustered experiments.

If two additive neighbors always receive the same cluster assignment W, then

$$Y_i = c_{i,\emptyset} + (c_{i,\{1\}} + c_{i,\{2\}})W.$$

The separate effects are indistinguishable, but the sum is the global contrast and is identified when $0 < \mathbb{P}_Z(W = 1) < 1$. The moment matrix based on $(1, Z_1, Z_2)$ is singular; its null direction changes the two coefficients by opposite amounts and leaves their sum unchanged. This is exactly the favorable case in Proposition 4.1.

More generally, under independent Bernoulli assignment of clusters, every configuration of the clusters touching a neighborhood has positive probability. If a polynomial vanishes on this support, it vanishes at both the all-treated and all-control cluster assignments, so its global contrast is zero. The desired contrast is therefore estimable even when the original unit-level coefficient vector is not.

Complete randomization of a fixed number of clusters can behave differently. For an extreme example, take a two-unit neighborhood and always treat exactly one of its members. The two additive schedules $Y_i(\mathbf{z}) = 0$ and $Y_i(\mathbf{z}) = z_1 + z_2 - 1$ agree at both supported assignments. Their all-treated-minus-all-control contrasts are 0 and 2. No estimator can distinguish them under this design. A restriction on total treatment count can remove precisely the variation needed to learn a global rollout effect.

Clustering also changes precision. It can reduce the number of independently varying treatment components relevant to an outcome, while correlating the weighted outcomes of more units. The cited work combines these effects in design-specific bounds and uses the bounds to compare candidate clusterings. The lesson is to choose the design and analysis together: a low-order model does not eliminate the reasons to cluster, and clustering does not eliminate the value of outcome structure.

8 Covariate adjustment should target the estimator's variance

The additive example showed that a large baseline outcome can create noise without changing the GATE. Suppose X_i is a fixed vector of pretreatment covariates predictive of that baseline. Since the first coordinate of Equation 13 gives $\mathbb{E}_Z[\ell_i] = 0$, any fixed coefficient vector θ yields the adjusted estimator

$$\hat{\tau}(\theta) = \frac{1}{N}\sum_i \ell_i(\mathbf{Z})\{Y_i^{\text{obs}} - X_i^\top \theta\}. \tag{18}$$

Its expectation is still τ_{G} whenever the original estimator is unbiased. The adjustment subtracts a random quantity with known mean zero. No correct linear model relating outcomes to X_i is required for this fixed-θ statement.

To see how to choose θ, define the random covariate vector $\mathbf{H} = N^{-1}\sum_i \ell_i X_i$, with $\mathbb{E}_Z[\mathbf{H}] = 0$. Then $\hat{\tau}(\theta) = \hat{\tau}(0) - \theta^\top \mathbf{H}$ and

$$\mathrm{Var}_Z\{\hat{\tau}(\theta)\} = \mathrm{Var}_Z\{\hat{\tau}(0)\} - 2\theta^\top \mathrm{Cov}_Z\{\mathbf{H}, \hat{\tau}(0)\} + \theta^\top \mathrm{Var}_Z(\mathbf{H})\theta. \tag{19}$$

If $\mathrm{Var}_Z(\mathbf{H})$ is invertible, the oracle choice is its inverse times $\mathrm{Cov}_Z\{\mathbf{H}, \hat{\tau}(0)\}$. This choice minimizes the variance of the final effect estimator. Ordinary regression of observed outcomes on covariates solves a different objective, and can neglect cross-unit covariance induced by interference.

Wang and Li (2026) make this difference central. Their variance-improvement-maximized (VIM) procedure estimates the change in variance from adjusting and chooses coefficients to maximize that improvement. A useful cancellation makes this feasible: the variance of the unadjusted estimator contains quadratic terms in unknown response coefficients, but those terms cancel when subtracting the adjusted variance. The remaining improvement can be estimated using the coefficient-estimation machinery already developed.

Their no-harm guarantee concerns the asymptotic variance of this particular procedure under the stated sparsity, moment, and coefficient-estimation conditions. It is not a guarantee for arbitrary regression adjustment or for every realized sample. Fixed coefficients preserve exact unbiasedness; fitting $\widehat{\theta}$ using the same experimental outcomes introduces additional randomness and requires an asymptotic argument. The paper also develops an asymptotically conservative variance estimator for inference.

In practice, choose covariates before examining treatment effects, use the dependence-aware adjustment procedure rather than substituting a familiar regression by analogy, and inspect whether the network conditions supporting its approximation are plausible. Randomly splitting outcome units does not automatically produce independent training and evaluation data, because their outcomes can depend on the same treatment coins.

9 Other structural information: monotone interference

9.1 Restricting directions instead of interaction order

An intervention may plausibly benefit from wider adoption without producing an additive response. A messaging feature might become useful only when a critical mass of friends uses it. A vaccination may become more protective as more neighbors are vaccinated. In such settings, the sign of a spillover can be more defensible than a low-order polynomial model.

One possible restriction is coordinatewise monotonicity: if $\mathbf{z} \leq \mathbf{z}'$ in every coordinate, then $Y_i(\mathbf{z}) \leq Y_i(\mathbf{z}')$. For spillovers alone, one might impose this ordering only while holding z_i fixed. The sign must match the outcome: beneficial vaccination could make disease incidence nonincreasing. These are substantive restrictions on potential outcomes, not consequences of randomization.

Monotonicity and low order are distinct. The function $\prod_{j\in\mathcal{N}_i} z_j$ is monotone and can have arbitrarily high order. An additive model with one positive and one negative coefficient is low order but not coordinatewise nondecreasing. Nor does monotonicity force every raw polynomial coefficient to be nonnegative: the function $z_1 + z_2 - z_1 z_2$, representing whether at least one person is treated, is monotone and has a negative pair coefficient.

9.2 What monotonicity can provide

For a simple illustration, define the marginal-treatment estimator

$$\hat{\tau}_{\text{marg}} = \frac{1}{N}\sum_i \left\{ \frac{Z_i Y_i^{\text{obs}}}{\pi_i} - \frac{(1-Z_i)Y_i^{\text{obs}}}{1-\pi_i} \right\}.$$

Its expectation compares the outcome conditional on own treatment to that conditional on own control. Under full coordinatewise nondecreasing outcomes, every outcome observed when $Z_i = 1$ is at most $Y_i(\mathbf{1}_N)$, and every outcome observed when $Z_i = 0$ is at least $Y_i(\mathbf{0}_N)$. Consequently,

$$\mathbb{E}_Z[\hat{\tau}_{\text{marg}}] \leq \tau_{\text{G}}. \tag{20}$$

This is a directional bias statement. It is neither a point-identification result for τ_{G} nor a claim that a realized estimate is a lower confidence bound. Establishing uncertainty around a bound needs further inference.

Several papers use structural inequalities for different purposes:

- Choi (2017) develops confidence intervals for attributable effects, comparing outcomes under the realized assignment with outcomes under no treatment. The restrictions resemble nonnegative treatment effects and do not require partial interference. That attributable-effect target differs from the all-treated-versus-all-control GATE.
- Eckles, Karrer, and Ugander (2017) use monotonicity and related conditions to study bias reduction through correlated assignment and exposure-based analysis, allowing potentially global interference. Their conditions explain when moving experimental exposure closer to a global policy comparison reduces bias; they do not make every clustering universally preferable.
- Pouget-Abadie and colleagues (2018) study comparing clustered designs when the corresponding estimators are biased in a common direction. If both expected estimates lie below the target, the larger expected estimate is closer to it. Their experiment-of-experiments design addresses how to compare clusterings experimentally. Here the relevant monotonicity is a property of bias across a family of designs, which should not be conflated with one generic unit-level inequality.
- Huang, Li, and Toulis (2025) develop finite-sample-valid randomization tests for monotone spillover contrasts using appropriate conditioning and network constructions. An inequality null does not reveal every missing potential outcome as a sharp null does; valid testing needs that additional construction. Such tests concern directional hypotheses, not unrestricted estimation of the GATE.

Submodularity is another possible restriction. Viewing an outcome as a function of the treated set, it says that the incremental effect of adding a given treated unit is no larger when the already-treated set is larger. It formalizes diminishing returns. It can coexist with monotonicity but is not implied by it, and it does not impose low polynomial order. The appropriate structure depends on the mechanism one is prepared to defend.

The central lesson of the lecture is that a partially treated network can inform a global rollout through several different kinds of structure. Low order turns the rollout contrast into a linear coefficient functional, and randomization supplies unbiased estimating equations for that functional.

Monotonicity instead supplies inequalities. In either case, the experiment's design determines what variation is available, and the structural assumption determines what that variation can tell us about unobserved assignments.

10 Advanced material: model error and the cost of extrapolation

10.1 An exact bias decomposition

Suppose a chosen order-β model leaves a remainder:

$$Y_i(\mathbf{z}) = \mathbf{v}_i(\mathbf{z})^\top \mathbf{c}_i + R_i(\mathbf{z}).$$

Assume the contrast is estimable for the retained polynomial space. The polynomial part remains unbiased, so the bias of the estimator for the actual GATE is exactly

$$\mathbb{E}_Z[\hat{\tau}_{\mathrm{PI}}] - \tau_{\mathrm{G}} = \frac{1}{N}\sum_i \left[\mathbb{E}_Z\{\ell_i(\mathbf{Z})R_i(\mathbf{Z})\} - \{R_i(\mathbf{1}_N) - R_i(\mathbf{0}_N)\}\right]. \tag{21}$$

Two features matter: how the omitted response aligns with the estimator's weight over the experimental distribution, and how much it contributes at the two policy endpoints. A model can fit typical experimental assignments well while misrepresenting the full rollout. With low treatment probability, a high-order interaction may rarely activate in the experiment yet contribute substantially when everyone is treated.

For example, if $|R_i(\mathbf{z})| \le \varepsilon_i$ uniformly over all assignments, Cauchy–Schwarz yields the bound

$$|\operatorname{Bias}_Z(\hat{\tau}_{\mathrm{PI}})| \le \frac{1}{N}\sum_i \varepsilon_i \left\{\sqrt{\mathbb{E}_Z[\ell_i^2]} + 2\right\}.$$

The uniform qualification is doing work: accuracy only on assignments that occur frequently does not control the endpoint remainder. Large weights also amplify model error over the experimental distribution.

If the contrast is not estimable under the design, there is an additional polynomial bias term $N^{-1}\sum_i \mathbf{a}_i^\top(\mathbf{P}_i - \mathbf{I})\mathbf{c}_i$, where $\mathbf{P}_i = \mathbf{M}_i^\dagger \mathbf{M}_i$ and $\mathbf{I}$ is the identity matrix. This separates lack of design support from misspecifying the outcome class. Either can obstruct the target, and they call for different responses.

10.2 Choosing an order and extending the experiment

Increasing β includes more response schedules but can increase the number and size of weights. A defensible analysis should connect the chosen order to a mechanism, examine sensitivity to plausible higher-order effects, and consider whether the design can improve the relevant treatment variation. Selecting the order after searching for a favorable estimated effect is not covered by the fixed-model unbiasedness argument.

The downstream literature also changes what data are collected. Under a common Bernoulli treatment probability p, the design mean of an order-β response is a univariate polynomial in p:

$$\frac{1}{N}\sum_i \mathbb{E}_{Z,p}[Y_i(\mathbf{Z})] = \frac{1}{N}\sum_i \sum_{S\in\mathcal{S}_i^\beta} c_{i,S}p^{|S|}.$$

Its value at one minus its value at zero is the GATE. Varying treatment saturation across rollout stages can therefore replace some detailed network information with observations at several values of p. But extrapolating a polynomial from a narrow range of probabilities to a full rollout can be unstable. Repeated stages also require a stable response interpretation, with time and carryover effects addressed by the design and assumptions.

Cortez-Rodriguez, Eichhorn, and Yu (2025) study a two-stage rollout that first selects a subpopulation and then rolls out treatment within it. Their analysis connects bias to cut edges and variance to clustering properties including covariate balance. Alongside clustered pseudoinverse estimation and VIM covariate adjustment, this is a concrete development beyond the original SNIPE construction: improve the information available to the estimator, rather than expecting an outcome restriction alone to solve every precision problem.

11 Further reading

Cortez-Rodriguez, Eichhorn, and Yu (2023) are the main source for this lecture. The centered-basis derivation above is a way to see the estimator's mechanics; the paper supplies the broader variance and asymptotic analysis. Eichhorn and colleagues (2026) extend the construction to general designs and make design selection part of the analysis. Their preprint first appeared in 2024; the cited version was revised in 2026.

For subsequent work, Wang and Li (2026) distinguish predicting outcomes well from reducing the variance of an interference estimator, while Cortez-Rodriguez and colleagues (2025) develop the rollout-design direction. These are complementary routes: use covariates, change dependence through clustering, or observe more than one treatment saturation.

The monotonicity papers cited above ask different causal questions with different inequalities. Reading them together is useful precisely because monotonicity is not one transferable assumption with one universal conclusion. Track the target, the direction of the restriction, the experimental design, and whether the result concerns bias, a bound, or a hypothesis test.

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